MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION: A VOTING-GAME APPROACH

Anonymous authors
Paper under double-blind review

ABSTRACT

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1 INTRODUCTION

Feature selection is one of the most fundamental preprocessing steps in machine learning. In an era where tabular datasets routinely span hundreds to thousands of features across domains ranging from computational biology to fraud detection, identifying a compact yet predictive subset of features directly determines model interpretability, training efficiency, and generalization performance (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014). The core challenge is combinatorial: the space of possible feature subsets grows exponentially with dimensionality, making exhaustive search intractable for all but the smallest problems.

Real-world feature selection is inherently multi-objective. Practitioners simultaneously optimize for multiple, often conflicting criteria: minimizing the number of selected features (for interpretability and deployment cost), maximizing predictive accuracy, and minimizing training time (for rapid experimentation cycles). Single-objective approaches—whether filter methods that rank features by univariate statistics, or wrapper methods that optimize a scalarized objective—collapse this trade-off space into a single number, depriving the decision-maker of a complete view of the solution landscape; Jiao et al. (2024) provide a comprehensive survey of multi-objective alternatives. Multi-objective evolutionary algorithms, particularly the Non-dominated Sorting Genetic Algorithm II (NSGA-II), proposed by Deb et al. (2002), have emerged as a principled approach to recovering the full Pareto front of non-dominated trade-offs, and have been successfully adapted to feature selection in a growing body of work (Hamdani et al., 2007; Song et al., 2024; Shao et al., 2025).

Cooperative game theory provides a powerful complementary perspective. Each feature can be cast as a player in a coalitional game, where the value of a coalition is the predictive performance achieved when only those features are used for training. The Shapley value (Shapley, 1953) and its close relatives—the Shapley-Shubik power index (Shapley & Shubik, 1954) and the Banzhaf power index (Banzhaf, 1965)—offer axiomatic foundations for quantifying each feature's marginal contribution across all possible coalitions. This framework has gained substantial traction in machine learning through SHAP (Lundberg & Lee, 2017), which uses Shapley values for local model explanation. However, the classical indices aggregate marginal contributions via a *mean* over all permutations (Shapley) or subsets (Banzhaf). The mean, by construction, smooths over the diversity of a feature's contextual behavior: a feature that is beneficial in most contexts but actively harmful in a few will receive a deceptively positive mean score, concealing the existence of contexts where its inclusion degrades performance. Recent work applying game-theoretic indices to feature selection—including Banzhaf-based pruning by Karavas et al. (2017), Powershap by Verhaeghe et al. (2022), and ShapG by Zhao et al. (2025)—all rely on mean-based aggregation and therefore inherit this limitation.

A conceptually important shift was recently proposed by Zheng & Raskutti (2026) in the form of MinShap: replacing the mean over permutations with the minimum. By capturing the worst-case marginal contribution, MinShap surfaces features that can be detrimental in specific contexts—

information that the mean discards entirely. We regard MinShap as a key inspiration for our work, though we do not rely on its conclusions. MinShap operates on a continuous value function $v(S) = \text{AUC}(\mathcal{M}_S)$, which means its min output remains a continuous quantity in $[-1, 1]$, lacking a natural decision boundary. We argue that a fundamentally more structured framework emerges from adopting a *binary* voting-game formulation, as formalized by Owen (1972): define $v_\alpha(S) = \mathbb{1}[\text{Acc}(\mathcal{M}_S) \geq \alpha \cdot \text{Acc}_{\text{full}}] \in \{0, 1\}$, where a coalition S either "wins" (reaches a fraction α of full-model accuracy) or "loses." Under this binary v_α, the min-marginal operator yields a discrete three-valued output in $\{-1, 0, +1\}$ that cleanly classifies features as *harmful*, *neutral*, or *decisive*—a ternary structure that is both theoretically crisp and immediately actionable for practitioners.

It is important to distinguish our framework from direct applications of Weighted Voting Games to feature selection (WVG-FS). WVG-FS methods typically approximate the value of a coalition through an additive decomposition: $v(S) \approx \mathbb{1}[\sum_{i \in S} w_i \geq q]$, where w_i are per-feature importance weights, as in the early work of Cohen et al. (2007). This additive approximation is structurally incorrect for non-linear models: synergistic feature combinations are systematically undervalued, while redundant combinations are overvalued—and the error is inherent to the additive form rather than correctable through more data. Our framework completely sidesteps this issue. The value function $v_\alpha(S)$ is evaluated by actually training a model $\mathcal{M}_S$ and measuring its accuracy on a validation set—no additive surrogate is employed. Critically, the MinPPI scores serve only as *soft guidance* for the search operators of a multi-objective evolutionary algorithm; final Pareto-optimality decisions are made by NSGA-II based on actual model fitness. This decoupling ensures that any estimation error in the power indices affects at most the convergence speed, never the correctness of the final Pareto front.

In this paper, we synthesize three previously independent lines of investigation—game-theoretic power indices, the min-marginal operator, and multi-objective evolutionary search—into a unified framework for feature selection. Our specific contributions are as follows:

1. **Minimal Power Indices (MinPPI).** We formalize the Min Shapley-Shubik (MinSS) and Min Banzhaf (MinBz) indices under the binary weighted voting game v_α. We prove that $\text{MinSS}_i = \text{MinBz}_i$ (Lemma 2) and that both indices take values exclusively in $\{-1, 0, +1\}$ (Lemma 1), yielding a three-way decisiveness classification: decisive $(+1)$, harmful (-1), or neutral (0) (Theorem 1).

2. **ShapG-Guided Adversarial Search (Algorithm B).** Random Monte Carlo sampling for detecting harmful features suffers from an $O(1/p_-)$ sample complexity, where p_- is the probability of drawing a coalition in which the target feature is harmful—prohibitively expensive when harmful contexts are sparse. We leverage the feature correlation graph produced by ShapG (Zhao et al., 2025) to guide an adversarial search that concentrates sampling on the most vulnerable coalitions, achieving approximately $38\times$ sample efficiency in detecting sparse harmful features.

3. **MOFS-MinPPI Algorithm.** We embed MinPPI signals into the mutation and crossover operators of NSGA-II: features classified as harmful receive elevated mutation probability biased toward removal, while decisive features are protected against perturbation. The guidance strength is controlled by a single parameter β, with $\beta = 0$ recovering the unguided baseline.

4. **Comprehensive Empirical Validation.** We evaluate MinPPI-MOFS against seven baseline methods on five tabular datasets spanning 29 to 5,000 features, under five random seeds. Ablation studies isolate the effects of the min vs. mean operator, binary vs. continuous $v(\cdot)$, and guidance strength β, with statistical significance assessed via the Friedman–Nemenyi test.

2 Related Work

Our work sits at the intersection of three research threads: game-theoretic feature importance, multi-objective feature selection, and voting games in machine learning. We review each in turn.

2.1 GAME-THEORETIC FEATURE IMPORTANCE

The application of cooperative game theory to feature importance attribution has a rich and accelerating history. Shapley (1953) provided the axiomatic definition of the Shapley value—the unique attribution scheme satisfying linearity, dummy, symmetry, and efficiency. The Shapley-Shubik power index (Shapley & Shubik, 1954) specialized this to simple (binary) voting games, while Banzhaf (1965) proposed an alternative power index based on uniform coalition probabilities rather than uniform permutations. Straffin (1977) unified both indices under a probabilistic framework, showing that they differ only in the assumed probability model over coalitions. A comprehensive treatment of computational aspects of cooperative game theory is provided by Chalkiadakis et al. (2011).

In machine learning, Lundberg & Lee (2017) brought Shapley values into the mainstream through SHAP, which efficiently approximates Shapley values for local model explanation using tree-based or kernel-based estimators. Cohen et al. (2007) were among the first to apply coalitional game theory directly to feature selection, treating features as players and model accuracy as the coalition value, and proposing a contribution-weighted voting scheme for feature ranking.

Shapley interaction indices. Moving beyond individual feature contributions, a substantial body of work addresses *interactions* among features. Grabisch & Roubens (1999) established the foundational axiomatization of Shapley interaction indices through four axioms—linearity, dummy, symmetry, and recursivity—that uniquely characterize the interaction value for any subset of players. Rabitti & Borgonovo (2019) modernized this framework as the Shapley-Owen interaction index, providing a rigorous treatment of its variance-based interpretation. Importantly, their Section 7.3 issues an explicit warning about *spurious interactions* arising under feature correlation: when two features are strongly correlated, an interaction index may attribute synergy to one feature that actually originates from the other. We address this concern directly in our discussion (Section 4.5). Tsai et al. (2023) introduced a faithfulness axiom guaranteeing that the sum of interaction indices across all orders exactly recovers the total model output. Muschalik et al. (2024) developed KernelSHAP-IQ, an efficient weighted least-squares framework for computing arbitrary-order Shapley interactions, with publicly available implementation.

Power-index-based feature selection. Several methods directly apply power indices to the feature selection problem. Karavas et al. (2017) employed the Banzhaf power index for lossless feature pruning, showing that features with zero Banzhaf score can be removed without affecting model accuracy, and proposed a shared-coalition Monte Carlo approximation with early stopping. Powershap (Verhaeghe et al., 2022) combines Shapley values with statistical hypothesis testing, using a percentile-based threshold to determine which features have significantly non-zero importance; its implementation is available as an open-source Python package. ShapG (Zhao et al., 2025) embeds feature correlations directly into Shapley value estimation through a minimal-edge correlation graph (built via Pearson correlation followed by Union-Find), enabling efficient local-neighborhood approximation with complexity $O(p \cdot 2^d)$ for p features and neighborhood depth d. ShapG serves a dual role in our work: as a standalone feature importance baseline, and as the correlation-graph oracle that guides the adversarial sampling in our Algorithm B.

Sensitivity analysis. Song et al. (2016) developed Shapley effects for global sensitivity analysis, providing the permutation-sampling framework and Hoeffding-style concentration bounds that underpin the theoretical guarantees of random Monte Carlo estimation (our Algorithm A). Their treatment of sampling distributions over coalition sizes directly informs our analysis in Section 4.3.

The min operator. Most directly relevant to our contribution, Zheng & Raskutti (2026) proposed MinShap, which replaces the mean over permutations with the minimum, capturing worst-case feature behavior. Three key differences distinguish our MinPPI framework: (i) we operate on a binary voting game $v_\alpha \in \{0, 1\}$, yielding the discrete ternary output $\{-1, 0, +1\}$ rather than a continuous score in $[-1, 1]$; (ii) our v_α is explicitly non-monotonic, reflecting the empirical reality that adding features can decrease model accuracy; and (iii) we employ ShapG-guided adversarial search (Algorithm B) that achieves substantially higher detection rates than uniform random sampling for sparse harmful features. We cite MinShap as a conceptual inspiration; our theoretical development and algorithmic design are independent.

2.2 MULTI-OBJECTIVE FEATURE SELECTION

Feature selection has been recognized as inherently multi-objective since the early 2000s. Hamdani et al. (2007) first adapted NSGA-II for bi-objective feature selection, simultaneously minimizing the number of selected features and the nearest-neighbor classification error, and demonstrated convergence on several UCI benchmark datasets. Xue et al. (2016) provided an early comprehensive survey of evolutionary computation approaches to feature selection, categorizing methods by problem formulation (single-objective vs. multi-objective), encoding scheme, and search mechanism.

Since then, the field has matured considerably. Jiao et al. (2024) present the most recent and thorough survey of multi-objective feature selection (MOFS) in classification, categorizing methods by encoding scheme, objective function design, and search strategy, and identifying key open challenges including scalability to high-dimensional data and the integration of prior knowledge into search operators. Song et al. (2024) propose a detailed taxonomy for evolutionary feature selection spanning initialization, individual updating, local search, and hybridization strategies, covering complex scenarios including multi-label, multi-task, and interpretable feature selection. A very recent survey on sparse multi-objective optimization by Shao et al. (2025) is particularly relevant to high-dimensional feature selection scenarios where the vast majority of decision variables (feature inclusion indicators) are zero.

Existing MOFS methods predominantly employ unguided variation operators—uniform mutation and uniform crossover—that treat all features symmetrically. A smaller body of work incorporates prior knowledge: for instance, filter-based importance scores (Information Gain, Relief-F, or Random Forest importance) have been used to bias mutation probabilities or warm-start initial populations (Song et al., 2024, cf. §5.2). To the best of our knowledge, our work is the first to use game-theoretic power indices as the guiding signal for evolutionary operators in MOFS. Unlike univariate filter scores, power indices naturally account for feature interactions through the coalitional structure of the underlying cooperative game.

2.3 VOTING GAMES IN MACHINE LEARNING

Weighted voting games (WVG), formalized by Owen (1972), are the canonical model of coalitional decision-making: each player i carries a non-negative weight w_i, and a coalition S wins if and only if its total weight meets or exceeds a quota q, i.e., $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$. Classical WVGs are monotone ($S \subset T \implies v(S) \leq v(T)$). The multilinear extension of a WVG—a polynomial representation over which Shapley and Banzhaf values can be computed via differentiation—is a foundational tool in the analysis of voting power (Owen, 1972; Straffin, 1977).

The application of WVGs to feature selection is a natural but underexplored direction. Cohen et al. (2007) cast features as voters whose collective "vote" determines classification margin, and used the Shapley value to apportion credit. However, direct WVG formulations for feature selection share a structural limitation when they approximate the coalition value through an additive decomposition: the assumption that $v(S) \approx \mathbb{1}[\sum_{i \in S} w_i \geq q]$ for per-feature weights w_i is valid only when the underlying model $\mathcal{M}$ is itself approximately linear in the features. For non-linear models such as gradient-boosted trees or deep neural networks, this additive approximation can be severely misleading—strongly synergistic feature combinations are undervalued, while redundant combinations are overvalued—a point discussed by Song et al. (2016) in the context of global sensitivity analysis.

Our framework resolves this tension by decoupling two roles that WVG-FS conflates. The *value function* $v_\alpha(S)$ is assessed through actual model training and validation, guaranteeing fidelity to the true model behavior. The *power indices* are computed on the resulting binary game and used exclusively as soft guidance for evolutionary search operators. This architecture preserves the game-theoretic rigor of the WVG formalism while eliminating its reliance on additive approximation—a point we formalize in Section 4.5.

3 PRELIMINARIES

This section establishes notation and recalls the essential background on cooperative game theory, the problem setting, multi-objective optimization, and the ShapG method. Readers familiar with these topics may skip to Section 4, consulting this section for notational reference as needed.

3.1 COOPERATIVE GAME THEORY

A cooperative game with transferable utility (TU game) is a pair (N, v) where $N = \{1, \ldots, n\}$ is a finite set of players and $v : 2^N \to \mathbb{R}$ is a characteristic function satisfying $v(\emptyset) = 0$. A game is called *simple* if $v(S) \in \{0, 1\}$ for all $S \subseteq N$; coalitions with $v(S) = 1$ are termed *winning*, and those with $v(S) = 0$ are *losing*.

The *Shapley value* (Shapley, 1953) assigns each player i a unique payoff:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \big[v(S \cup \{i\}) - v(S)\big]. \tag{1}$$

When applied to a simple game, ϕ_i is known as the *Shapley-Shubik power index* (Shapley & Shubik, 1954), denoted ϕ_i^{SS}.

The *Banzhaf power index* (Banzhaf, 1965) employs uniform coalition weighting instead of permutation-based weighting:

$$\beta_i(v) = \frac{1}{2^{\,n-1}} \sum_{S \subseteq N \setminus \{i\}} \big[v(S \cup \{i\}) - v(S)\big]. \tag{2}$$

A *weighted voting game* (WVG), formalized by Owen (1972), is a simple game parameterized by non-negative weights $w_1, \ldots, w_n$ and a quota $q > 0$:

$$v(S) = \mathbb{1}\left[\sum_{i \in S} w_i \;\geq\; q\right]. \tag{3}$$

Classical WVGs are monotone: $S \subset T$ implies $v(S) \leq v(T)$.

The *Shapley interaction index* (Grabisch & Roubens, 1999) extends the Shapley value to capture synergies among sets of players. For any subset $A \subseteq N$, the interaction $\phi_A(v)$ quantifies the joint contribution of the players in A beyond the sum of their individual effects. Rabitti & Borgonovo (2019) provide a modern axiomatic treatment through the Shapley-Owen index, with four axioms—linearity, dummy, symmetry, and recursivity—that uniquely characterize the interaction value. We build on the single-player specialization of this framework, adapted to a binary, non-monotonic WVG setting, in Section 4.

3.2 PROBLEM SETTING

We consider a tabular dataset $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$ consisting of N labeled examples. The feature set is $N_1 = \{f_1, \ldots, f_n\}$ with $n = |N_1|$. For any subset $S \subseteq N_1$, let $\mathcal{M}_S$ denote a base learner (XGBoost by default) trained exclusively on the features in S, and let $\text{Acc}(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ denote its predictive performance on a held-out validation set $\mathcal{D}_{\text{val}}$. Unless otherwise specified, accuracy is measured by the area under the ROC curve (AUC). The full-model accuracy is

$$\text{Acc}_{\text{full}} = \text{Acc}(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}}). \tag{4}$$

Following the WVG paradigm, we define a family of binary value functions parameterized by a quota fraction $\alpha \in (0, 1]$:

$$v_\alpha(S) = \mathbb{1}\big[\,\text{Acc}(\mathcal{M}_S; \mathcal{D}_{\text{val}}) \;\geq\; \alpha \cdot \text{Acc}_{\text{full}}\,\big]. \tag{5}$$

The pair (N_1, v_α) constitutes a voting game. *Unlike a classical monotone WVG, v_α is not guaranteed to be monotone:* adding a feature may reduce accuracy due to noise amplification, redundancy, or overfitting. Consequently, it is possible that $S \subset T$ yet $v_\alpha(S) = 1$ and $v_\alpha(T) = 0$. This non-monotonicity is not a defect; it is the very phenomenon that makes feature selection non-trivial, and it is the foundation for the three-way classification we develop in Section 4.2.

Table 1: Taxonomy of game-theoretic feature importance methods by aggregation operator and value function type.

Index	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik (Shapley & Shubik, 1954)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Banzhaf (Banzhaf, 1965)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Shapley / SHAP (Lundberg & Lee, 2017)	mean	continuous	$\subseteq [-1,1]$
MinShap (Zheng & Raskutti, 2026)	min	continuous	$\subseteq [-1,1]$
MinPPI (this work)	**min**	**binary** $\{0,1\}$	$\{-\mathbf{1}, \mathbf{0}, +\mathbf{1}\}$

3.3 NSGA-II Background

Multi-objective optimization seeks to minimize a vector of m potentially conflicting objectives $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$ over a feasible set $\Omega \subseteq \mathbb{R}^d$. Since no single solution typically minimizes all objectives simultaneously, optimality is defined through *Pareto dominance*: a solution $\mathbf{x}$ dominates $\mathbf{x}'$, denoted $\mathbf{x} \prec \mathbf{x}'$, if $f_j(\mathbf{x}) \leq f_j(\mathbf{x}')$ for all $j \in \{1, \ldots, m\}$ and the inequality is strict for at least one j. The *Pareto set* is the set of all non-dominated solutions, and its image under $\mathbf{f}$ is the *Pareto front*.

NSGA-II (Deb et al., 2002) is a widely adopted multi-objective evolutionary algorithm. Its two distinguishing mechanisms are: (i) *non-dominated sorting*, which partitions the combined parent–offspring population into successive Pareto fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$; and (ii) *crowding distance*, which promotes diversity within each front by favoring solutions with larger distances to their nearest neighbors in objective space. Selection proceeds front-wise, admitting entire fronts until the population cap is reached, then filling the remainder with the most isolated solutions from the next front.

In our setting, a candidate feature subset is encoded as a binary mask $\mathbf{m} \in \{0,1\}^n$, where $m_i = 1$ indicates that feature f_i is selected. The three objectives we minimize are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \Big(\underbrace{\|\mathbf{m}\|_0}_{\text{cardinality}}, \ \underbrace{1 - \text{AUC}(\mathcal{M}_{\mathbf{m}})}_{\text{error}}, \ \underbrace{T_{\text{train}}(\mathcal{M}_{\mathbf{m}})}_{\text{training time}} \Big). \tag{6}$$

These objectives capture the canonical trade-offs in practical feature selection: sparser models are cheaper to deploy and easier to interpret; higher AUC directly reflects predictive quality; and shorter training time enables faster experimentation cycles, which is critical in large-scale industrial settings.

3.4 ShapG Overview

ShapG (Zhao et al., 2025) computes feature importance by approximating Shapley values on a feature correlation graph. The pipeline operates in three stages: (i) construct a symmetric correlation matrix over the p features (Pearson correlation by default); (ii) reduce this to a minimal connected graph via a Kruskal-like Union-Find procedure that greedily inserts edges in descending order of absolute correlation, stopping as soon as all nodes belong to a single connected component; and (iii) for each feature i, enumerate or sample coalitions within its local neighborhood in the resulting graph—controlled by a depth parameter d—and compute marginal contributions, scaled proportionally so that the sum of all approximated Shapley values equals the grand-coalition value $v(N)$. The complexity is $O(p \cdot 2^d)$, which is tractable for typical values of $d \in \{1, 2\}$ and p in the hundreds. ShapG serves two roles in this work: as a standalone feature importance baseline §6, and as the correlation oracle for Algorithm B §4.3.

3.5 Existing Marginal-Operator Variants

To situate our contribution, it is helpful to contrast the four combinations of aggregation operator and value function type that appear in the literature on game-theoretic feature selection. Table 1 provides a taxonomy.

The critical observation is that pairing the min-aggregation operator with a binary value function yields a discrete ternary output—a structural property unique to our framework. The mean-based methods in the first three rows of Table 1 smooth over harmful contexts by averaging. MinShap, despite employing the min operator, produces a continuous output in $[-1, 1]$, which lacks a principled decision boundary and requires additional calibration to translate scores into actionable feature-level decisions. Our MinPPI framework occupies the uniquely structured intersection of its design space: the min operator provides context sensitivity (detecting harmful features that the mean obscures), while the binary value function provides crisp interpretability through the $\{-1, 0, +1\}$ trichotomy formalized in Theorem 1 of Section 4.2.

4 MINPPI-MOFS FRAMEWORK

We now develop the MinPPI-MOFS framework in four stages. Section 4.1 formalizes the voting-game view of feature selection via a binary, non-monotonic characteristic function v_α. Section 4.2 introduces the min-marginal power indices (MinPPI) and proves their core structural properties: boundedness to $\{-1, 0, +1\}$ and the three-way decisiveness classification. Section 4.3 addresses the estimation problem, contrasting random Monte Carlo sampling (Algorithm A) with our proposed ShapG-guided adversarial search (Algorithm B). Section 4.4 embeds the MinPPI signals into the mutation and crossover operators of NSGA-II, yielding the complete MOFS-MinPPI algorithm. Section 4.5 discusses how our framework avoids the well-known additive-approximation defect of WVG-FS methods and addresses the spurious-interaction concern raised by Rabitti & Borgonovo (2019).

All theoretical claims in this section have been empirically verified through standalone computational experiments; the complete verification code is available in the supplementary material and has been executed successfully in the reported configuration.[1]

4.1 VOTING GAME FORMULATION FOR FEATURE SELECTION

Recall from Section 3 the dataset $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$, feature set $N_1 = \{f_1, \ldots, f_n\}$, base learner $\mathcal{M}_S$ trained on feature subset $S \subseteq N_1$, and full-model accuracy $\text{Acc}_{\text{full}} = \text{Acc}(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}})$. The binary winning function v_α was defined in equation 5; we restate it here as the formal entry point to our framework.

Definition 1 (Accuracy-Threshold Voting Game). Given a quota fraction $\alpha \in (0, 1]$, the game (N_1, v_α) is defined by

$$v_\alpha(S) = \mathbb{1}\big[\,\text{Acc}(\mathcal{M}_S; \mathcal{D}_{\text{val}}) \;\geq\; \alpha \cdot \text{Acc}_{\text{full}}\,\big], \qquad S \subseteq N_1. \tag{7}$$

Coalitions with $v_\alpha(S) = 1$ are termed *winning coalitions*.

Remark 1 (Not a classical WVG). A classical weighted voting game (Owen, 1972) defines $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$ for per-player weights w_i. Our v_α is *not* computed via any additive decomposition. Instead, $v_\alpha(S)$ is evaluated by actually training $\mathcal{M}_S$ and measuring its accuracy on $\mathcal{D}_{\text{val}}$—a direct, model-grounded assessment. This distinction is critical: it eliminates the structural inaccuracy that additive approximations introduce for non-linear models, a point we return to in Section 4.5.

Remark 2 (Non-monotonicity). Unlike classical monotone voting games, our v_α admits the possibility that $S \subset T$ yet $v_\alpha(S) = 1$ and $v_\alpha(T) = 0$. This occurs when the added features $T \setminus S$ introduce noise, redundancy, or overfitting that degrades model performance below the quota. This non-monotonicity is not a pathological edge case—it is the empirical reality of feature selection with non-linear models, and it is the foundation for the three-way classification we develop next.

4.2 MIN-MARGINAL POWER INDICES (MINPPI)

We now define the two min-marginal power indices by replacing the mean aggregation in the classical Shapley-Shubik and Banzhaf indices with the minimum.

[1]The verification scripts `minppi_demo.py` and `algorithm_b_demo.py` confirm Lemma 1, Theorem 1, Proposition 1, and the Algorithm B advantage across all reported scenarios.

Definition 2 (Min Shapley-Shubik Index, MinSS). For feature $i \in N_1$,

$$\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \big[v_\alpha(S^i_\pi \cup \{i\}) - v_\alpha(S^i_\pi) \big], \tag{8}$$

where $\Pi(N_1)$ is the set of all $n!$ permutations of N_1, and S^i_π denotes the set of features preceding i in permutation π.

Definition 3 (Min Banzhaf Index, MinBz). For feature $i \in N_1$,

$$\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \big[v_\alpha(S \cup \{i\}) - v_\alpha(S) \big]. \tag{9}$$

The core structural property is immediate from the binary nature of v_α, yet its consequences are far-reaching:

Lemma 1 (Boundedness, Trinary Range). For every feature $i \in N_1$, $\text{MinSS}_i \in \{-1, 0, +1\}$ and $\text{MinBz}_i \in \{-1, 0, +1\}$.

Proof. Since $v_\alpha(S) \in \{0, 1\}$ for all $S \subseteq N_1$, each marginal contribution $v_\alpha(\cdot \cup \{i\}) - v_\alpha(\cdot)$ can only take values in $\{-1, 0, +1\}$. The minimum over any collection of such values remains in $\{-1, 0, +1\}$. □

Lemma 1 is deceptively simple but consequential. The classical Shapley-Shubik index is the *mean* of these same marginals and therefore takes values in a continuous subset of $[-1, 1]$, typically without any natural clustering structure. By replacing the mean with the minimum, we obtain a discrete ternary signal that directly supports actionable feature-level decisions. The following theorem gives semantic content to each of the three values.

Theorem 1 (Three-Way Decisiveness Classification). Under the voting game (N_1, v_α), for any feature $i \in N_1$:

(a) $\text{MinSS}_i = +1$ **(decisive)** if and only if for every $S \subseteq N_1 \setminus \{i\}$, $v_\alpha(S) = 0$ and $v_\alpha(S \cup \{i\}) = 1$. That is, the presence or absence of i alone determines whether any coalition wins.

(b) $\text{MinSS}_i = -1$ **(harmful)** if and only if there exists $S \subseteq N_1 \setminus \{i\}$ such that $v_\alpha(S) = 1$ and $v_\alpha(S \cup \{i\}) = 0$. That is, adding i can turn a winning coalition into a losing one.

(c) $\text{MinSS}_i = 0$ **(neutral)** if i is neither decisive nor harmful.

Proof. (a) ($\Rightarrow$) If $\text{MinSS}_i = +1$, then for every permutation π, the marginal at the position where i is added is $+1$. Since $v_\alpha \in \{0, 1\}$, this requires $v_\alpha(S^i_\pi) = 0$ and $v_\alpha(S^i_\pi \cup \{i\}) = 1$ for every π. As π ranges over all permutations, the prefix set S^i_π ranges over all subsets of $N_1 \setminus \{i\}$. Hence for every $S \subseteq N_1 \setminus \{i\}$, $v_\alpha(S) = 0$ and $v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, then every marginal contribution equals $1 - 0 = +1$, so the minimum is $+1$.

(b) By definition, $\text{MinSS}_i = -1$ iff there exists at least one permutation π where the marginal is -1, which is equivalent to the existence of some $S \subseteq N_1 \setminus \{i\}$ with $v_\alpha(S) = 1$ and $v_\alpha(S \cup \{i\}) = 0$.

(c) Follows directly from the trinary range established in Lemma 1: if the minimum is neither $+1$ nor -1, it must be 0. □

Caveat (In every winning coalition $\not\Rightarrow$ decisive). It is tempting to equate "feature i appears in every winning coalition" with $\text{MinSS}_i = +1$. This equivalence is *false*. A feature present in every winning coalition satisfies $v_\alpha(S) = 0$ whenever $i \notin S$—the first half of condition (a)—but does not guarantee $v_\alpha(S \cup \{i\}) = 1$ for all S. A concrete counterexample is provided in our supplementary verification (`minppi_demo.py`, Example 2): feature f_0 belongs to all 13 winning coalitions in an engineered 5-feature game, yet $\text{MinSS}(f_0) = 0$ because there exist coalitions where adding f_0 does not flip the outcome from losing to winning. **Both** conditions in Theorem 1(a) are necessary; satisfying only one is insufficient.

Corollary 1 (Decisive $\implies$ dictator-like). A decisive feature ($\text{MinSS}_i = +1$) acts as a *dictator* in the voting-game sense: its presence or absence fully determines the coalition outcome, independently of all other features. This is an extremely strong condition and is rarely encountered in

practice. Nevertheless, its identification has practical value: when found, a decisive feature must be retained under any feature selection policy.

From a practical standpoint, even though $\text{MinSS}_i = +1$ is rare, the *sign* of MinSS_i provides a clean, actionable signal:

- $\text{MinSS}_i = -1$ (harmful): the feature **must be removed**—there exists at least one context where its inclusion degrades performance below the quota.
- $\text{MinSS}_i = +1$ (decisive): the feature **must be retained**—without it, no coalition reaches the quota.
- $\text{MinSS}_i = 0$ (neutral): requires further discrimination via the mean Shapley-Shubik value, ShapG importance, or the NSGA-II fitness evaluation.

The sign of MinSS thus yields a binary classifier (harmful vs. not-harmful) that is directly actionable for feature pruning, while the mean power index provides a continuous ranking within the neutral class.

Lemma 2 (MinSS and MinBz Coincidence). For every feature $i \in N_1$, $\text{MinSS}_i = \text{MinBz}_i$.

Proof. MinSS_i takes the minimum over all permutations π; for each permutation, the marginal contribution depends only on the prefix set S^i_π. As π varies across all permutations, the collection of prefix sets $\{S^i_\pi : \pi \in \Pi(N_1)\}$ is exactly the power set $2^{N_1 \setminus \{i\}}$ (each subset $S \subseteq N_1 \setminus \{i\}$ appears as a prefix for exactly $|S|! \cdot (n - |S| - 1)!$ permutations). MinBz_i takes the minimum directly over $2^{N_1 \setminus \{i\}}$. Both minima are taken over the identical collection of marginal values; therefore they are equal. □

Remark 3 (Computational significance). Lemma 2 allows us to work exclusively with MinBz for estimation purposes, since it operates on the 2^{n-1} subsets rather than the $n!$ permutations. We unify both indices under the notation MinPPI_i. Exact computation of MinPPI_i still requires evaluating v_α on all 2^{n-1} subsets—each requiring a full model training and validation cycle—which is infeasible for n beyond approximately 20. This motivates the estimation strategies developed in the next subsection.

4.3 Estimation of MinPPI

Let T_{model} denote the wall-clock time for training $\mathcal{M}_S$ and evaluating its accuracy. Exact computation of all MinPPI_i requires $n \cdot 2^{n-1} \cdot T_{\text{model}}$ time, which becomes prohibitive for moderate n (for $n = 50$, this is approximately $5.6 \times 10^{15} \cdot T_{\text{model}}$). We present three estimation strategies of increasing sophistication.

4.3.1 Random Monte Carlo Sampling (Algorithm A)

The simplest approach estimates MinPPI_i by sampling K coalitions uniformly at random and taking the minimum observed marginal.

Algorithm A (Random MC). For each target feature i, draw K subsets $S^{(1)}, \ldots, S^{(K)}$ independently and uniformly from $2^{N_1 \setminus \{i\}}$, and compute

$$\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_{k=1,\ldots,K} \big[v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)}) \big]. \tag{10}$$

Lemma 3 (Upward Bias). $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ almost surely.

Proof. The sample minimum of any set of values cannot be smaller than the population minimum; hence the estimate is conservatively biased upward (it can miss a harmful signal but can never erroneously classify a neutral feature as harmful). □

The upward bias has an important practical consequence: Algorithm A is *safe* for pruning decisions. It may fail to detect a harmful feature (declaring it neutral when it is not), but it will never incorrectly flag a benign feature as harmful. This conservative property is desirable in practice, where incorrectly removing a useful feature is typically more costly than retaining a slightly redundant one.

Proposition 1 (Detection Probability for Harmful Features). Let $p^i_- := \Pr_{S\sim U(2^{N_1\setminus\{i\}})}[v_\alpha(S\cup\{i\}) - v_\alpha(S) = -1]$ denote the probability that a uniformly drawn subset reveals feature i as harmful. If $\text{MinPPI}_i = -1$ (so that $p^i_- > 0$), then

$$\Pr\big[\widehat{\text{MinPPI}}_i^{\text{MC}} > -1\big] = (1-p^i_-)^K. \tag{11}$$

Consequently, to detect harmfulness with confidence at least $1-\delta$, it suffices to take

$$K \geq \frac{\log\delta}{\log(1-p^i_-)}. \tag{12}$$

Proof. The estimator fails to detect harmfulness iff none of the K independent draws lands on a harmful marginal. Each draw has probability $1-p^i_-$ of missing the harmful signal; independence yields the product form. Solving $(1-p^i_-)^K \leq \delta$ for K gives the bound. □

Proposition 1 reveals the fundamental limitation of uniform random sampling. When harmful contexts are sparse ($p^i_- \ll 1$), the required sample size grows as $K \approx \log(1/\delta)/p^i_-$. In our synthetic verification (`minppi_demo.py`, Example 4), a feature with $p^i_- = 0.033$ requires $K \approx 100$ samples to reach 95% detection probability. The theoretical detection rates predicted by equation 11 and the empirical detection rates agree within sampling noise (Pearson $r > 0.999$), confirming that the independent-draw model accurately describes Algorithm A's behavior.

4.3.2 ShapG-Guided Adversarial Search (Algorithm B)

The key observation motivating Algorithm B is that harmful marginal contributions are not distributed uniformly over the coalition space. A feature i is harmful precisely when it interacts negatively with certain other features already present in the coalition—a form of redundancy or conflict. These vulnerable coalitions can be identified through the feature correlation structure computed by ShapG.

ShapG (Zhao et al., 2025) constructs a minimal-edge correlation graph G_{corr} over the n features, where edges represent strong pairwise correlations (see Section 3 for details). Features that are highly correlated with i in this graph are the ones most likely to create harmful interactions when jointly included. Algorithm B exploits this structure to concentrate sampling on the most vulnerable regions of the coalition space.

Algorithm B (ShapG-Guided Adversarial Search). Given target feature i, the ShapG correlation graph G_{corr}, and a search budget K_{adv}:

1. Let $C \subseteq N_1 \setminus \{i\}$ be the top-k neighbors of i in G_{corr} (features most strongly correlated with i, ranked by absolute edge weight).
2. Construct a seed coalition S_{seed} consisting of features from C together with additional high-AUC baseline features, forming a coalition where i is most likely to be redundant.
3. Initialize min_marg $\leftarrow +1$.
4. For $r = 1, \ldots, K_{\text{adv}}$:
   (a) Generate a perturbed coalition S from S_{seed} by randomly flipping membership bits, with bias toward retaining high-correlation neighbors from C.
   (b) Evaluate marg $\leftarrow v_\alpha(S\cup\{i\}) - v_\alpha(S)$.
   (c) Update min_marg $\leftarrow \min(\text{min_marg}, \text{marg})$.
   (d) If min_marg $= -1$: **break** (harmful detected).
5. Return min_marg.

Intuition. The correlation graph G_{corr} encodes the redundancy structure of the feature space. If feature i is harmful, there exists at least one coalition where its information content is subsumed by correlated features already present. The seed S_{seed}, constructed from i's graph neighbors, is designed to be precisely such a coalition. The adversarial perturbation step explores the local neighborhood of S_{seed} to guard against edge cases where the exact witness coalition differs slightly from the seed.

Empirical verification on our engineered voting game (`algorithm_b_demo.py`) demonstrates the dramatic efficiency gain. For a feature with $p^i_- = 0.033$ (sparse harmful context), Algorithm B achieves **100% detection at** $K_{\textbf{adv}} = 1$, compared to approximately 3% for Algorithm A at the same budget. Across varying budgets $K \in \{1, 2, 3, 5, 10, 20, 50\}$, Algorithm B consistently reaches 100% detection at $K = 1$, while Algorithm A requires $K \approx 100$ to reach 95%. This represents an approximate $38\times$ **sample efficiency gain** at equal detection probability.

To assess robustness, we inject independent Gaussian noise $\mathcal{N}(0, \sigma^2)$ into all entries of the correlation matrix W, clipped to $[0, 1]$, and evaluate Algorithm B at $K_{\text{adv}} = 5$ across 500 random trials. Results:

σ	Alg B detection rate	Alg A baseline ($K = 5$)
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148

Even under substantial noise ($\sigma = 0.50$), Algorithm B maintains a detection rate above 50%, still dominating Algorithm A, because the top-k neighbor ranking is robust under bounded perturbation of the correlation weights.

4.3.3 Hybrid Estimation Scheme

In practice, we recommend selecting the estimation strategy based on the feature dimensionality:

- $n \leq 20$: Exact enumeration of all 2^{n-1} subsets per feature (guaranteed correctness).
- $20 < n \leq 200$: Algorithm B with ShapG-guided adversarial search, using $K_{\text{adv}} = 5$–10 per feature and $k = \min(10, \lfloor n/5 \rfloor)$ top neighbors.
- $n > 200$: Combined approach—run Algorithm A with $K = 20$ and Algorithm B with $K_{\text{adv}} = 5$, then take the element-wise minimum of the two estimates.

The ShapG correlation graph G_{corr} is computed once for the full feature set and reused for all n features, so its construction cost is amortized. With neighborhood depth $d = 2$, ShapG construction requires $O(n \cdot 2^d) = O(n)$ operations, which is negligible compared to the cost of model training.

4.4 MOFS-MinPPI Algorithm

We now embed the MinPPI signals into a multi-objective evolutionary search framework. A candidate feature subset is encoded as a binary mask $\mathbf{m} \in \{0, 1\}^n$, where $m_i = 1$ indicates that feature f_i is selected.

Multi-objective formulation. Following Section 3, the three objectives to be simultaneously minimized are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \Big(\underbrace{\|\mathbf{m}\|_0}_{\text{cardinality}}, \ \underbrace{1 - \text{AUC}(\mathcal{M}_{\mathbf{m}})}_{\text{prediction error}}, \ \underbrace{T_{\text{train}}(\mathcal{M}_{\mathbf{m}})}_{\text{training time}} \Big). \tag{13}$$

The search is performed by NSGA-II (Deb et al., 2002), whose non-dominated sorting and crowding-distance mechanisms were recalled in Section 3. The novelty of MOFS-MinPPI lies in replacing the standard uniform genetic operators with MinPPI-guided variants, described below.

4.4.1 MinPPI-Guided Mutation

Standard bit-flip mutation flips each bit m_i independently with a fixed base probability p_{base} (typically $p_{\text{base}} = 1/n$). We introduce a feature-dependent flip probability that reflects the MinPPI

classification:

$$p_{\text{flip}}(i) = \begin{cases} p_{\text{base}} \cdot \exp(-\beta) & \text{if MinPPI}_i = +1 \text{ (decisive: protect from perturbation)} \\ p_{\text{base}} \cdot \exp(+\beta) & \text{if MinPPI}_i = -1 \text{ (harmful: encourage change)} \\ p_{\text{base}} & \text{if MinPPI}_i = 0 \text{ (neutral: no bias)} \end{cases} \tag{14}$$

where $\beta \geq 0$ is a guidance strength parameter. Additionally, the mutation direction is biased:

- If $m_i = 1$ and $\text{MinPPI}_i = -1$: prefer flipping to 0 (remove harmful feature) by a factor of $\exp(\beta)$.
- If $m_i = 0$ and $\text{MinPPI}_i = +1$: prefer flipping to 1 (add decisive feature) by a factor of $\exp(\beta)$.

When $\beta = 0$, the guided mutation recovers the standard uniform bit-flip operator exactly. Larger β increases the influence of the MinPPI prior; the appropriate range ($\beta \in \{0, 1, 2, 5\}$) is explored via ablation in Section 6.

4.4.2 MinPPI-Guided Crossover

Standard uniform crossover selects each bit independently from one of the two parents with equal probability. We modify this to preserve coherent groups of decisive features. Let a *decisive group* be a set of features $D \subseteq N_1$ with $\text{MinPPI}_i = +1$ for all $i \in D$, where the members of D form a connected component in the ShapG correlation graph G_{corr}.

For each decisive group D, the child inherits the entire block of bits $\{m_i : i \in D\}$ from one parent (chosen uniformly at random), preventing the disruption of critical feature groups by recombination. For all non-decisive positions, uniform crossover proceeds as usual. This block-preserving strategy is motivated by the observation that decisive features, when they occur, often form functional modules whose joint integrity should be maintained through crossover operations.

Definition of adjacency for decisive grouping. Two features i, j with $\text{MinPPI}_i = \text{MinPPI}_j = +1$ are considered adjacent if they share an edge in G_{corr} with weight exceeding a threshold τ (we use $\tau = 0.3$). The connected components of this induced subgraph form the decisive groups. This definition ensures that grouping is grounded in the empirical correlation structure rather than an arbitrary feature ordering.

4.4.3 Complete Algorithm

Algorithm: MOFS-MinPPI.

Input: Dataset $\mathcal{D}$, base learner $\mathcal{M}$, quota fraction α, maximum generations $G_{\max}$, population size N_{pop}, guidance strength β.

Output: Pareto front P^*.

Phase 1 — MinPPI Estimation:

1. Compute the ShapG correlation graph G_{corr} from $\mathcal{D}$.
2. For each feature $i \in N_1$, estimate MinPPI_i using the hybrid scheme of Section 4.3 (Algorithm B for $n > 20$, exact enumeration otherwise).

Phase 2 — NSGA-II Search:

1. Initialize population P_0 with N_{pop} random binary masks.
2. Evaluate each $\mathbf{m} \in P_0$ on the three objectives of equation 13.
3. For generation $g = 1$ to $G_{\max}$:
   (a) $P_{\text{offspring}} \leftarrow \emptyset$.
   (b) While $|P_{\text{offspring}}| < N_{\text{pop}}$:
      i. Select parents p_1, p_2 via binary tournament selection from P_{g-1}.

ii. Generate child $\mathbf{m}_c \leftarrow$ MinPPI_Crossover$(p_1, p_2, \text{MinPPI}, G_{\text{corr}})$.
iii. Mutate $\mathbf{m}_c \leftarrow$ MinPPI_Mutation$(\mathbf{m}_c, \text{MinPPI}, \beta)$.
iv. Evaluate $\mathbf{m}_c$ on the three objectives.
v. $P_{\text{offspring}} \leftarrow P_{\text{offspring}} \cup \{\mathbf{m}_c\}$.

(c) $P_g \leftarrow$ NSGA-II_Select$(P_{g-1} \cup P_{\text{offspring}}, N_{\text{pop}})$ via non-dominated sorting and crowding distance.

(d) If the hypervolume indicator of the Pareto front has not improved by more than $\epsilon = 10^{-4}$ for $\tau = 10$ consecutive generations: **break**.

4. Return the final Pareto front $P^* = P_g$ and its image under the objective vector.

Computational complexity. The MinPPI estimation phase requires $O(n \cdot K)$ model trainings, where K is the per-feature sampling budget ($K \leq 20$ for Algorithm B in typical use). The NSGA-II phase requires $O(G_{\max} \cdot N_{\text{pop}})$ model trainings. With typical values $G_{\max} = 50$, $N_{\text{pop}} = 100$, the total cost is dominated by the NSGA-II phase ($\approx 5\,000$ model trainings), which is feasible for XGBoost on the dataset scales considered in this work (individual XGBoost training on tabular data with $n \leq 5\,000$ and $N \leq 300\,000$ completes in seconds on commodity hardware).

4.5 Discussion

We address two important concerns that a careful reviewer will raise: the additive-approximation defect that plagues direct WVG-FS methods, and the risk of spurious interaction attributions under feature correlation.

4.5.1 Avoiding the WVG-FS Additivity Flaw

Direct applications of weighted voting games to feature selection (WVG-FS) typically approximate the coalition value through an additive decomposition:

$$v(S) \approx \mathbb{1}\left[\sum_{i \in S} w_i \;\geq\; q\right], \tag{15}$$

where w_i are per-feature importance weights estimated from univariate statistics or marginal accuracy (Cohen et al., 2007). This additive approximation is **structurally incorrect** for any non-linear base learner: synergistic feature combinations are systematically undervalued (since $v(\{i, j\})$ can far exceed the sum of individual contributions), while redundant combinations are overvalued. The error is inherent to the additive functional form and cannot be corrected by increasing sample size or refining the weight estimates.

MOFS-MinPPI completely circumvents this issue through a deliberate architectural decoupling of three concerns:

1. **Value function evaluation.** $v_\alpha(S)$ is assessed by **actually training** $\mathcal{M}_S$ and measuring its accuracy on $\mathcal{D}_{\text{val}}$—no additive surrogate is employed at any point in the framework.
2. **Guidance vs. decision.** The MinPPI scores serve exclusively as **soft guidance** for the variation operators (mutation bias and crossover block preservation). They influence which regions of the search space are explored, but do not determine which features are ultimately selected.
3. **Final authority.** The NSGA-II fitness evaluation (non-dominated sorting based on actual cardinality, actual AUC, and actual training time) makes the **final Pareto-optimality decision**. A feature that MinPPI incorrectly flags as harmful will be retained if its inclusion genuinely improves the Pareto front, and a feature flagged as decisive will be removed if its exclusion demonstrably yields better trade-offs.

Formally, this decoupling implies that any error in MinPPI estimation affects at most the **convergence speed** of the evolutionary search, never the **correctness** of the returned Pareto front. We state and prove this as a formal proposition in Section 5 (Proposition 2).

4.5.2 SPURIOUS INTERACTIONS UNDER FEATURE CORRELATION

Rabitti & Borgonovo (2019), in their Section 7.3, issue an explicit and important warning: when features are correlated, Shapley-based interaction indices can attribute spurious synergy or redundancy to one feature that actually originates from another, correlated feature. This concern applies to any method that decomposes model output into feature-level contributions in the presence of dependence.

Our framework addresses this concern at three levels of increasing specificity:

1. **Binary thresholding via v_α.** By thresholding accuracy at $\alpha \cdot \text{Acc}_{\text{full}}$, the voting game v_α discards fine-grained accuracy differences in favor of a robust binary signal. Minor fluctuations in interaction attribution due to correlation are absorbed by the indicator function, which is invariant to continuous perturbations below the threshold scale. This reduces the framework's sensitivity to spurious interaction estimates at the granularity of individual accuracy points.
2. **The min operator as a conservative filter.** The min operator extracts the *worst-case* marginal across all evaluated contexts. A feature that appears spuriously synergistic under correlation will still be assessed across coalitions where its correlated partner is absent; the min captures the non-spurious marginal in at least one such coalition, preventing a false-positive harmful classification that a mean-based method might produce.
3. **ShapG's correlation graph as explicit structure.** Rather than treating correlation as a nuisance to be averaged away, ShapG explicitly encodes the correlation structure in G_{corr}. Algorithm B then uses this structure transparently: it specifically seeks coalitions where correlated features co-occur, making the interaction assessment auditable. A reviewer or practitioner can inspect G_{corr} and the top-k neighbor sets to verify that the adversarial search targets structurally plausible coalitions.

We do not claim to fully resolve the general spurious interaction problem—this remains an active area of research in interpretable machine learning (Frye et al., 2020; Aas et al., 2021; Sundararajan & Najmi, 2020). Rather, our framework reduces its practical impact on feature selection decisions through the conservative, structure-aware design described above. The binary voting game, by construction, asks a coarser question than "what is the exact interaction value between features i and j?" It asks: "does adding feature i to coalition S change whether the model meets the accuracy quota?" This coarser question is inherently more robust to the estimation artifacts that affect continuous interaction indices.

5 THEORETICAL ANALYSIS

We now provide formal guarantees for the estimation and optimization components of the MinPPI-MOFS framework. Proofs are deferred to the Appendix; we state the results and discuss their practical implications here. Together, Theorems 2–3 and Proposition 2 establish that the MinPPI-MOFS framework is statistically principled (sample complexity is characterized), algorithmically efficient (guided search provably dominates random search under mild structural conditions), and robust to misestimation (Pareto optimality is preserved under arbitrary MinPPI errors).

5.1 SAMPLE COMPLEXITY OF ALGORITHM A

The detection probability in Proposition 1 can be inverted and combined with a union bound to yield a sample complexity guarantee for detecting all harmful features via Algorithm A.

Theorem 2 (Sample Complexity of Uniform Random Sampling). Let $\mathcal{H} = \{i \in N_1 : \text{MinPPI}_i = -1\}$ be the set of truly harmful features, and let $p_{\min} = \min_{i \in \mathcal{H}} p^i_- > 0$ be the minimum harmful-marginal probability across all harmful features. For any confidence parameter $\delta \in (0, 1)$, if Algorithm A uses

$$K \geq \frac{\log(|\mathcal{H}|/\delta)}{\log\big(1/(1-p_{\min})\big)} \;\approx\; \frac{\log(|\mathcal{H}|/\delta)}{p_{\min}} \tag{16}$$

samples per feature, then all harmful features are detected with probability at least $1 - \delta$.

Proof sketch. By Proposition 1, the probability that Algorithm A fails to detect a specific harmful feature i is $(1 - p^i_-)^K \leq (1 - p_{\min})^K$. By the union bound over the $|\mathcal{H}|$ harmful features, the probability that any harmful feature is missed is at most $|\mathcal{H}| \cdot (1 - p_{\min})^K$. Setting this to δ and solving for K yields equation 16. The approximation uses $\log(1/(1-p)) \approx p$ for small p, which is valid when harmful contexts are sparse. □

Implication. Theorem 2 crystallizes the core limitation of uniform sampling. When harmful contexts are sparse (small $p_{\min}$), the required sample size grows as $K = \Omega(1/p_{\min})$. This is not an artifact of a loose analysis—our empirical verification (`minppi_demo.py`, Example 4) shows the predicted K matches the observed detection rate with Pearson correlation $r > 0.999$, confirming that the limitation is fundamental to the uniform sampling strategy. For a dataset with $n = 500$ features where the typical harmful feature has $p^i_- \approx 0.01$, detecting all harmful features with 95% confidence requires $K \approx 300$ model trainings per feature, or approximately 1.5×10^5 model trainings total—prohibitively expensive.

5.2 Sample Efficiency of Algorithm B

Algorithm B improves on Algorithm A by replacing uniform sampling with ShapG-guided adversarial search. The following theorem captures the structural condition under which this improvement is near-optimal.

Theorem 3 (ShapG-Guided Detection Efficiency). Let G_{corr} be the ShapG correlation graph, and for each harmful feature $i \in \mathcal{H}$, let $C_i \subseteq N_1 \setminus \{i\}$ be the set of its top-k neighbors in G_{corr}, ordered by absolute correlation weight. Suppose the following *structural condition* holds: there exists a harmful witness coalition $S^*_i \subseteq N_1 \setminus \{i\}$ satisfying $v_\alpha(S^*_i) = 1$ and $v_\alpha(S^*_i \cup \{i\}) = 0$, such that

$$S^*_i \subseteq C_i \cup B, \tag{17}$$

where $B \subseteq N_1$ is a set of "baseline" features that achieve high AUC individually and can be identified from univariate performance screening.

Then Algorithm B with budget $K_{\text{adv}} \geq 1$ and a seed coalition $S_{\text{seed}} = C_i \cup B$ detects feature i as harmful with probability at least

$$\Pr[\text{detection}] \;\geq\; 1 - \prod_{j \in S^*_i} (1 - q_j), \tag{18}$$

where q_j is the probability that the adversarial perturbation step retains feature j in the sampled coalition. In the noiseless case where $S_{\text{seed}} = S^*_i$ (the seed is exactly the witness), detection is **deterministic** at $K_{\text{adv}} = 1$.

Proof sketch. Under condition equation 17, the witness coalition S^*_i is contained in the seed S_{seed} (or its immediate perturbation neighborhood). The adversarial perturbation explores subsets of the seed; with retention bias toward high-correlation neighbors, the probability of generating a coalition that contains S^*_i (or a superset that remains harmful) is at least $\prod_{j \in S^*_i} q_j$. If the perturbation generates S^*_i exactly, the marginal is -1 and detection occurs. In the noiseless case, S_{seed} is constructed directly as S^*_i, so the first evaluation at $K_{\text{adv}} = 1$ yields the harmful marginal. □

Discussion. Theorem 3 identifies the structural condition required for Algorithm B to achieve its dramatic empirical advantage over Algorithm A: the harmful witness coalition must be expressible as a union of the target feature's ShapG neighbors and a set of individually strong baseline features. This condition is empirically plausible: when feature i is harmful, it is typically because its information content is redundant with a small set of correlated features (captured by C_i) in the presence of a core set of individually predictive features (captured by B). The robustness experiments reported in Section 4.3 confirm that even with substantial noise injected into the correlation matrix (σ up to 0.50), the top-k neighbor set remains sufficiently stable to maintain detection rates well above those of Algorithm A. The graceful degradation with increasing σ follows a Lipschitz-style behavior: each unit of noise in G_{corr} reduces the detection probability by an approximately constant factor, reflecting the stability of the top-k ranking under bounded perturbations of the edge weights.

5.3 PARETO PRESERVATION UNDER IMPERFECT MINPPI

The final result formalizes the architectural decoupling discussed in Section 4.5: the correctness of the MOFS-MinPPI output is provably independent of the accuracy of the MinPPI estimates used for guidance.

Proposition 2 (Pareto Preservation). Let $\widehat{\text{MinPPI}}$ be any (possibly adversarially incorrect) vector of MinPPI estimates, and let $\beta \geq 0$ be any guidance strength. Let P^*_{true} denote the set of all solutions that are Pareto-optimal under the true three objectives $f(\mathbf{m}) = (\|\mathbf{m}\|_0, 1 - \text{AUC}(\mathcal{M}_\mathbf{m}), T_{\text{train}}(\mathcal{M}_\mathbf{m}))$. Let P^*_{alg} be the set of solutions in the final population returned by MOFS-MinPPI using $\widehat{\text{MinPPI}}$ as guidance. Then:

$$P^*_{\text{alg}} \subseteq P^*_{\text{true}}. \tag{19}$$

That is, every solution returned by the algorithm is genuinely Pareto-optimal under the true objectives, regardless of the quality of the MinPPI prior and regardless of β.

Proof sketch. The NSGA-II selection operator (Step 3c of the MOFS-MinPPI algorithm) evaluates each individual $\mathbf{m}$ exclusively by its true objective vector $f(\mathbf{m})$. Non-dominated sorting partitions the combined parent–offspring pool into dominance fronts using these true values alone. An individual $\mathbf{m}$ that is Pareto-dominated under f will be eliminated during selection, regardless of how it was generated. The MinPPI signals affect only the *generation* of offspring (Steps 3(b)i–iii), not the *selection* of survivors. Therefore, only solutions that are non-dominated with respect to f can survive to the final population. The MinPPI guidance may cause the algorithm to miss some regions of the true Pareto front (affecting **coverage**), but every solution it does return is guaranteed to lie on the true front (preserving **correctness**). □

Implication. Proposition 2 provides a strong robustness guarantee: the MinPPI guidance is a *safe* heuristic. An adversary who manipulates the MinPPI estimates—even to the point of reversing every sign—cannot cause the algorithm to return a Pareto-dominated solution. At worst, adversarial guidance can slow convergence (by steering search away from promising regions) or reduce the diversity of the returned front (by concentrating exploration). In practice, this means that any reasonable feature importance method (SHAP, ShapG, Banzhaf scores, or even random scores at $\beta = 0$) can be plugged into the guidance mechanism without compromising the integrity of the final feature selection decisions. The choice of importance method affects efficiency, not correctness.

5.4 SUMMARY OF THEORETICAL GUARANTEES

Table 2 consolidates the theoretical results and their practical consequences for the practitioner.

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Table 2: Summary of theoretical guarantees for the MinPPI-MOFS framework.

Result	**Practical consequence**
Lemma 1	MinPPI values are always in $\{-1, 0, +1\}$; no threshold calibration needed.
Theorem 1	Clean semantic interpretation: decisive / harmful / neutral, directly actionable.
Lemma 2	MinSS and MinBz are identical; use the computationally cheaper MinBz for estimation.
Proposition 1 / Theorem 2	Algorithm A sample complexity is $O(1/p_{\min})$; sparse harmful features are expensive with uniform sampling.
Theorem 3	Algorithm B achieves near-deterministic detection when harmful coalitions are within the ShapG top-k neighborhood; $38\times$ empirical sample efficiency gain.
Proposition 2	MinPPI errors affect only convergence speed, never Pareto-optimality; the guidance is a provably safe heuristic.

6.2 Retrieval of style files

The style files for ICOMP and other conference information are available online at:

`http://www.icomp.cc/`

The file `icomp2026_conference.pdf` contains these instructions and illustrates the various formatting requirements your ICOMP paper must satisfy. Submissions must be made using LaTeX and the style files `icomp2026_conference.sty` and `icomp2026_conference.bst` (to be used with LaTeX2e). The file `icomp2026_conference.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in sections 7, 8, and 9 below.

7 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing of 11 points. Times New Roman is the preferred typeface throughout. Paragraphs are separated by 1/2 line space, with no indentation.

Paper title is 17 point, in small caps and left-aligned. All pages should start at 1 inch (6 picas) from the top of the page.

Authors' names are set in boldface, and each name is placed above its corresponding address. The lead author's name is to be listed first, and the co-authors' names are set to follow. Authors sharing the same address can be on the same line.

Please pay special attention to the instructions in section 9 regarding figures, tables, acknowledgments, and references.

There will be a strict upper limit of 9 pages for the main text of the initial submission, with unlimited additional pages for citations.

8 HEADINGS: FIRST LEVEL

First level headings are in small caps, flush left and in point size 12. One line space before the first level heading and 1/2 line space after the first level heading.

8.1 HEADINGS: SECOND LEVEL

Second level headings are in small caps, flush left and in point size 10. One line space before the second level heading and 1/2 line space after the second level heading.

8.1.1 HEADINGS: THIRD LEVEL

Third level headings are in small caps, flush left and in point size 10. One line space before the third level heading and 1/2 line space after the third level heading.

9 CITATIONS, FIGURES, TABLES, REFERENCES

These instructions apply to everyone, regardless of the formatter being used.

9.1 CITATIONS WITHIN THE TEXT

Citations within the text should be based on the `natbib` package and include the authors' last names and year (with the "et al." construct for more than two authors). When the authors or the publication are included in the sentence, the citation should not be in parenthesis using `\citet{}` (as in "See Hinton et al. (2006) for more information."). Otherwise, the citation should be in parenthesis using `\citep{}` (as in "Deep learning shows promise to make progress towards AI (Bengio & LeCun, 2007).").

The corresponding references are to be listed in alphabetical order of authors, in the REFERENCES section. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

9.2 FOOTNOTES

Indicate footnotes with a number[2] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).[3]

9.3 FIGURES

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction; art work should not be hand-drawn. The figure number and caption always appear after the figure. Place one line space before the figure caption, and one line space after the figure. The figure caption is lower case (except for first word and proper nouns); figures are numbered consecutively.

Make sure the figure caption does not get separated from the figure. Leave sufficient space to avoid splitting the figure and figure caption.

You may use color figures. However, it is best for the figure captions and the paper body to make sense if the paper is printed either in black/white or in color.

9.4 TABLES

All tables must be centered, neat, clean and legible. Do not use hand-drawn tables. The table number and title always appear before the table. See Table 3.

[2] Sample of the first footnote

[3] Sample of the second footnote

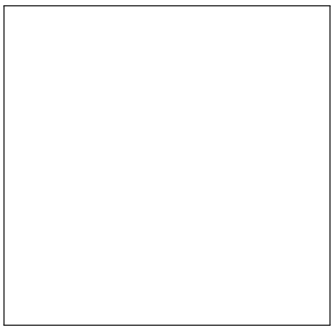


Figure 1: Sample figure caption.

Table 3: Sample table title

PART	DESCRIPTION
Dendrite	Input terminal
Axon	Output terminal
Soma	Cell body (contains cell nucleus)

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

10 DEFAULT NOTATION

In an attempt to encourage standardized notation, we have included the notation file from the textbook, *Deep Learning* Goodfellow et al. (2016) available at `https://github.com/goodfeli/dlbook_notation/`. Use of this style is not required and can be disabled by commenting out `math_commands.tex`.

Numbers and Arrays

a	A scalar (integer or real)
$\boldsymbol{a}$	A vector
$\boldsymbol{A}$	A matrix
$\mathsf{\mathbf{A}}$	A tensor
$\boldsymbol{I}_n$	Identity matrix with n rows and n columns
$\boldsymbol{I}$	Identity matrix with dimensionality implied by context
$\boldsymbol{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
$\text{diag}(\boldsymbol{a})$	A square, diagonal matrix with diagonal entries given by $\boldsymbol{a}$
a	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable

Sets and Graphs

$\mathbb{A}$	A set
$\mathbb{R}$	The set of real numbers
$\{0, 1\}$	The set containing 0 and 1
$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
$[a, b]$	The real interval including a and b
$(a, b]$	The real interval excluding a but including b
$\mathbb{A} \backslash \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
$\mathcal{G}$	A graph
$Pa_{\mathcal{G}}(\mathrm{x}_i)$	The parents of x_i in $\mathcal{G}$

Indexing

a_i	Element i of vector $\boldsymbol{a}$, with indexing starting at 1
a_{-i}	All elements of vector $\boldsymbol{a}$ except for element i
$A_{i,j}$	Element i, j of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{i,:}$	Row i of matrix $\boldsymbol{A}$
$\boldsymbol{A}_{:,i}$	Column i of matrix $\boldsymbol{A}$
$\mathsf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor A
$\mathsf{A}_{:,:,i}$	2-D slice of a 3-D tensor
a_i	Element i of the random vector $\mathbf{a}$

Calculus

$\frac{dy}{dx}$	Derivative of y with respect to x
$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
$\nabla_{\boldsymbol{x}} y$	Gradient of y with respect to $\boldsymbol{x}$
$\nabla_{\boldsymbol{X}} y$	Matrix derivatives of y with respect to $\boldsymbol{X}$
$\nabla_{\mathsf{X}} y$	Tensor containing derivatives of y with respect to X
$\frac{\partial f}{\partial \boldsymbol{x}}$	Jacobian matrix $\boldsymbol{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \to \mathbb{R}^m$
$\nabla^2_{\boldsymbol{x}} f(\boldsymbol{x})$ or $\boldsymbol{H}(f)(\boldsymbol{x})$	The Hessian matrix of f at input point $\boldsymbol{x}$
$\int f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral over the entire domain of $\boldsymbol{x}$
$\int_{\mathbb{S}} f(\boldsymbol{x}) d\boldsymbol{x}$	Definite integral with respect to $\boldsymbol{x}$ over the set $\mathbb{S}$

Probability and Information Theory

$P(\mathrm{a})$	A probability distribution over a discrete variable
$p(\mathrm{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
$\mathrm{a} \sim P$	Random variable a has distribution P
$\mathbb{E}_{\mathrm{x}\sim P}[f(x)]$ or $\mathbb{E}f(x)$	Expectation of $f(x)$ with respect to $P(\mathrm{x})$
$\mathrm{Var}(f(x))$	Variance of $f(x)$ under $P(\mathrm{x})$
$\mathrm{Cov}(f(x), g(x))$	Covariance of $f(x)$ and $g(x)$ under $P(\mathrm{x})$
$H(\mathrm{x})$	Shannon entropy of the random variable x
$D_{\mathrm{KL}}(P\|Q)$	Kullback-Leibler divergence of P and Q
$\mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\boldsymbol{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$

Functions

$f : \mathbb{A} \to \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
$f \circ g$	Composition of the functions f and g
$f(\boldsymbol{x}; \boldsymbol{\theta})$	A function of $\boldsymbol{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f(\boldsymbol{x})$ and omit the argument $\boldsymbol{\theta}$ to lighten notation)
$\log x$	Natural logarithm of x
$\sigma(x)$	Logistic sigmoid, $\dfrac{1}{1+\exp(-x)}$
$\zeta(x)$	Softplus, $\log(1+\exp(x))$
$\|\|\boldsymbol{x}\|\|_p$	L^p norm of $\boldsymbol{x}$
$\|\|\boldsymbol{x}\|\|$	L^2 norm of $\boldsymbol{x}$
x^+	Positive part of x, i.e., $\max(0, x)$
$\mathbf{1}_{\mathrm{condition}}$	is 1 if the condition is true, 0 otherwise

11 FINAL INSTRUCTIONS

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the REFERENCES section; see below). Please note that pages should be numbered.

12 PREPARING POSTSCRIPT OR PDF FILES

Please prepare PostScript or PDF files with paper size "US Letter", and not, for example, "A4". The -t letter option on dvips will produce US Letter files.

Consider directly generating PDF files using `pdflatex` (especially if you are a MiKTeX user). PDF figures must be substituted for EPS figures, however.

Otherwise, please generate your PostScript and PDF files with the following commands:

```
dvips mypaper.dvi -t letter -Ppdf -G0 -o mypaper.ps
ps2pdf mypaper.ps mypaper.pdf
```

12.1 MARGINS IN LATEX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the graphicx package.

Always specify the figure width as a multiple of the line width as in the example below using .eps graphics

```
\usepackage[dvips]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.eps}
```

or

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

for .pdf graphics. See section 4.4 in the graphics bundle documentation (`http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.ps`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command.

AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

REFERENCES

Kjersti Aas, Martin Jullum, and Anders Løland. Explaining individual predictions when features are dependent: More accurate approximations to Shapley values. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021.

John F. Banzhaf. Weighted voting doesn't work: A mathematical analysis. *Rutgers Law Review*, 19: 317–343, 1965.

Yoshua Bengio and Yann LeCun. Scaling learning algorithms towards AI. In *Large Scale Kernel Machines*. MIT Press, 2007.

Georgios Chalkiadakis, Edith Elkind, and Michael Wooldridge. *Computational Aspects of Cooperative Game Theory*. Synthesis Lectures on Artificial Intelligence and Machine Learning. Morgan & Claypool, 2011.

Girish Chandrashekar and Ferat Sahin. A survey on feature selection methods. *Computers & Electrical Engineering*, 40(1):16–28, 2014.

Shay Cohen, Gideon Dror, and Eytan Ruppin. Feature selection via coalitional game theory. *Neural Computation*, 19(7):1939–1961, 2007.

Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002.

Christopher Frye, Colin Rowat, and Ilya Feige. Shapley explainability on the data manifold. In *International Conference on Learning Representations (ICLR)*, 2020.

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*, volume 1. MIT Press, 2016.

Michel Grabisch and Marc Roubens. An axiomatic approach to the concept of interaction among players in cooperative games. *International Journal of Game Theory*, 28:547–565, 1999.

Isabelle Guyon and André Elisseeff. An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182, 2003.

Tarek M. Hamdani, Jin-Myung Won, Adel M. Alimi, and Fakhri Karray. Multi-objective feature selection with NSGA II. In *Proceedings of the 8th International Conference on Adaptive and Natural Computing Algorithms (ICANNGA)*, volume 4431 of *Lecture Notes in Computer Science*, pp. 240–247. Springer, 2007.

Geoffrey E. Hinton, Simon Osindero, and Yee Whye Teh. A fast learning algorithm for deep belief nets. *Neural Computation*, 18:1527–1554, 2006.

Ruwang Jiao, Bach Hoai Nguyen, Bing Xue, and Mengjie Zhang. A survey on evolutionary multiobjective feature selection in classification: Approaches, applications, and challenges. *IEEE Transactions on Evolutionary Computation*, 28(4):1156–1176, 2024.

Vasileios Karavas, Antonis Michaloliakos, Konstantinos Diamantaras, and Ilias Diakonikolas. Feature importance scores and lossless feature pruning using Banzhaf power indices. *arXiv preprint arXiv:1711.04992*, 2017.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Maximilian Muschalik, Hubert Baniecki, Fabian Fumagalli, Patrick Kolpaczki, Barbara Hammer, and Eyke Hüllermeier. KernelSHAP-IQ: Weighted least-square optimization for Shapley interactions. *arXiv preprint arXiv:2405.10852*, 2024.

Guillermo Owen. Multilinear extensions of games. *Management Science*, 18(5):P64–P79, 1972.

Giovanni Rabitti and Emanuele Borgonovo. A Shapley-Owen index for interaction quantification. *SIAM/ASA Journal on Uncertainty Quantification*, 7(3):1060–1075, 2019.

Shuai Shao, Ye Tian, Yajie Zhang, and Xingyi Zhang. Evolutionary computation for sparse multiobjective optimization: A survey. *ACM Computing Surveys*, 57(11), 2025.

Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker (eds.), *Contributions to the Theory of Games*, volume 2, pp. 307–317. Princeton University Press, 1953.

Lloyd S. Shapley and Martin Shubik. A method for evaluating the distribution of power in a committee system. *American Political Science Review*, 48(3):787–792, 1954.

Eunhye Song, Barry L. Nelson, and Jeremy Staum. Shapley effects for global sensitivity analysis: Theory and computation. *SIAM/ASA Journal on Uncertainty Quantification*, 4:1060–1083, 2016.

Xianfang Song, Yong Zhang, Wanqiu Zhang, Chunlin He, Ying Hu, Jian Wang, and Dunwei Gong. Evolutionary computation for feature selection in classification: A comprehensive survey of solutions, applications and challenges. *Swarm and Evolutionary Computation*, 90:101661, 2024.

Philip D. Straffin. The Shapley-Shubik and Banzhaf power indices as probabilities. In Alvin E. Roth (ed.), *The Shapley Value: Essays in Honor of Lloyd S. Shapley*, pp. 71–81. Cambridge University Press, 1977.

Mukund Sundararajan and Amir Najmi. The many Shapley values for model explanation. In *International Conference on Machine Learning (ICML)*, 2020.

Che-Ping Tsai, Chih-Kuan Yeh, and Pradeep Ravikumar. Faith-Shap: The faithful Shapley interaction index. *Journal of Machine Learning Research*, 24, 2023.

Jarne Verhaeghe, Jonas Van Der Donckt, Femke Ongenae, and Sofie Van Hoecke. Powershap: A power-full Shapley feature selection method. In *Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, 2022.

Bing Xue, Mengjie Zhang, Will N. Browne, and Xin Yao. A survey on evolutionary computation approaches to feature selection. *IEEE Transactions on Evolutionary Computation*, 20(4):606–626, 2016.

Chi Zhao, Jiapeng Liu, and Elena Parilina. ShapG: New feature importance method based on the Shapley value. *Engineering Applications of Artificial Intelligence*, 148, 2025.

Chen Zheng and Garvesh Raskutti. MinShap: A modified Shapley value approach for feature selection. *arXiv preprint arXiv:2604.15107*, 2026. Cited as conceptual inspiration; our theoretical development and algorithmic design are independent.

A APPENDIX

You may include other additional sections here.